

Project Title

The Mini Project report submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Technology

 $\dot{in}$

Computer Science & Engineering

A.B. Alpha (Reg. No. SYALECS001)

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Sixth Semester 2011 Admission



Sreepathy Institute of Management & Technology

Vavanoor, Palakkad-679533

Affiliated to

University Of Calicut

Department of Computer Science & Engineering
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CERTIFICATE

This is to certify that the report entitled "**Name Of The Project**" is a bonafide record of the Mini Project presented by Name 1 (Reg. No. SYALECS001), Name 2 (Reg. No. SYALECS001), **Name 3 (Reg. No. SYALECS001)** and Name 4 (Reg. No. SYALECS001) under our supervision and guidance. The report has been submitted to the Department of Computer Science & Engineering of SIMAT Vavanoor, Palakkad-679533 in partial fulfillment of the award of the Degree of Bachelor of Technology in Computer Science & Engineering, during the year 2013-2014.

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ABSTRACT

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Chapter 1

Introduction

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1.1 Problem definition

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1.2 Chapter details

The report is organised as follows: In order to fully understand the background of keyphrase extraction, chapter two presents the related works which includes a literature review of the extraction mechanisms so far developed. Chapter three gives an overview of neural network and backpropagation learning method. From chapter four onwards, different stages of the proposed keyphrase extraction method are described with suitable block diagrams and algorithm. Chapter four has the document preprocessing, chapter five, the candidate phrase identification and chapter six, the different features used for describing phrases. Chapter seven clearly describes how training and testing is being done in the neural network. Chapter eight gives the testing strategies and results obtained and chapter nine the summary of obtained results. The last chapter, ten, give the conclusion and futureworks.

Chapter 2

Requirement Analysis

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2.1 Existing System

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2.2 Proposed System

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Chapter 3

Module Description

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Chapter 4

Module 1 Name

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The proposed framework for efficient data deduplication is shown below.

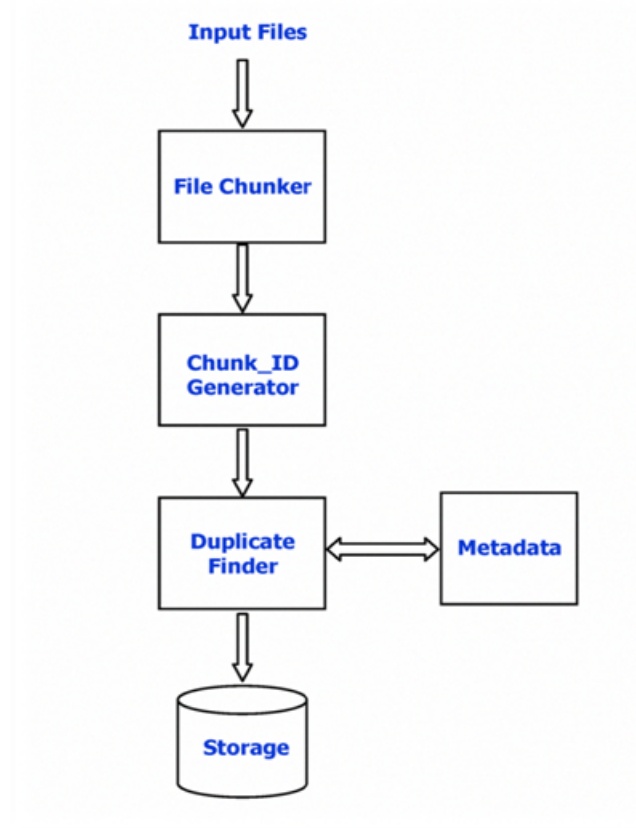


Figure 4.1: Architecture

Chapter 5

Module 2 Name

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5.1 Insertion

Insert takes as input a NDB and string x in $\{0,1\}^l$ and outputs a NDB' that matches string matched by NDB and every string matched by x . The purpose of the insert operation is to introduce a subset of strings into the negative database while safeguarding its irreversibility properties. Lines 1 and 2 enable the procedure to create several entries in NDB portraying x . Likewise steps 3 and 4 set some of the unspecified positions of x (if any) so that it may be possible for a set of strings representing x , that exhibit bits not found in x , to be entered in NDB. Finally the call to Negative Pattern Generate produces y that represents x ; y is then inserted in NDB.

Insert(x ,NDB)

1. Randomly choose $1 \leq j \leq l$
2. for $k = 1$ to j do
3. Randomly select from x at most n distinct unspecified bit positions
4. for every possible bit assignment B_p of the selected positions
5. $x^j \leftarrow x \times B_p$

6. $y \leftarrow \text{Negative} - \text{Pattern} - \text{Generate}\{x^l, NDB\}$
7. add y to NDB.

Negative-Pattern-Generate(x, NDB)

1. Create a random permutation π .
2. for all specified bits bi in $\pi(x)$,
3. Let x' be the same as $\pi(x)$ but with bi flipped.
4. if x' is subsumed by some string in $\pi(NDB)$,
5. $\pi(x) \leftarrow \pi(x)$ -ith bit (set value to *)
6. Keep track of the i th bit in a set indicator vector (SIV).
7. Randomly choose $0 \leq t \leq \|SIV\|$
8. $R \leftarrow t$ randomly selected bits from SIV.
9. Create a pattern V_k using $\pi(x)$ and the bits indicated by R .
10. Return $\pi'(V_k)$ where π' is the inverse permutation of π

Negative-Pattern-Generate take as input a string x defined over $0, 1, *$ and a database NDB and outputs a string that matches x and nothing else outside of NDB.

The morph operation [1] takes as input a NDB and outputs a NDB' that matches exactly the same set of binary strings; however, NDB and NDB' are different, that means some NDB records are not in NDB' and vice-versa. This illustrates the ability of the negative database scheme to have different representations for the same data. A property that makes it difficult to determine if two negative databases are equivalent (match exactly the same set of binary strings) without reversing them. Uses of the Morph operation include removing superfluous entries from the negative database and obscuring the history of operations performed on it.

5.2 Application Of Operators

Imagine a goods company that wishes to keep a list of risky credit-card numbers: credit-cards that have been involved in suspicious activities. A hard-to-reverse negative database can be created such that the ability to retrieve individual numbers is hindered, preventing a malicious party from stealing any numbers, and the capability to validate specific numbers is preserved. Suppose that the department that holds the negative database discovers that some number in the list is not delinquent. The Insert operation allows that party to remove the entry without finding out anything about any other record in the database. Likewise, if a new number comes along that is regarded as suspicious, it can be safely included in the watch list via the Delete operation.

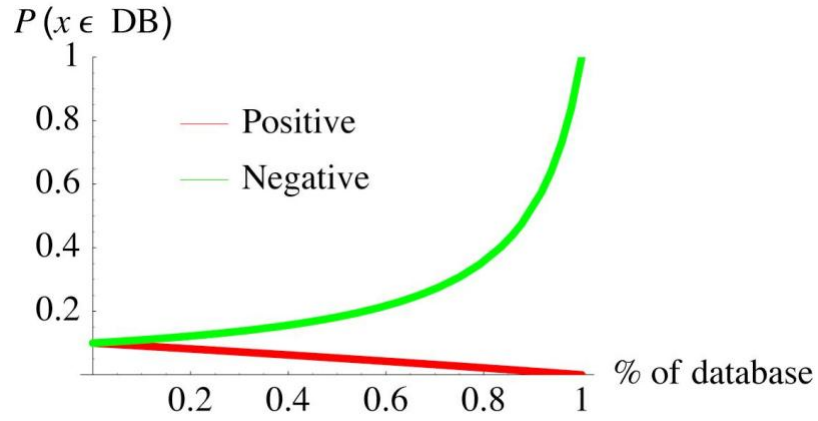


Figure 5.1: DB contains 10% of all possible L-length strings (formulas)

Table 5.1: Nonlinear Model Results

Case	Method#1	Method#2	Method#3
1	50	837	970
2	47	877	230
3	31	25	415
4	35	144	2356
5	45	300	556

Chapter 6

Data Flow Diagram

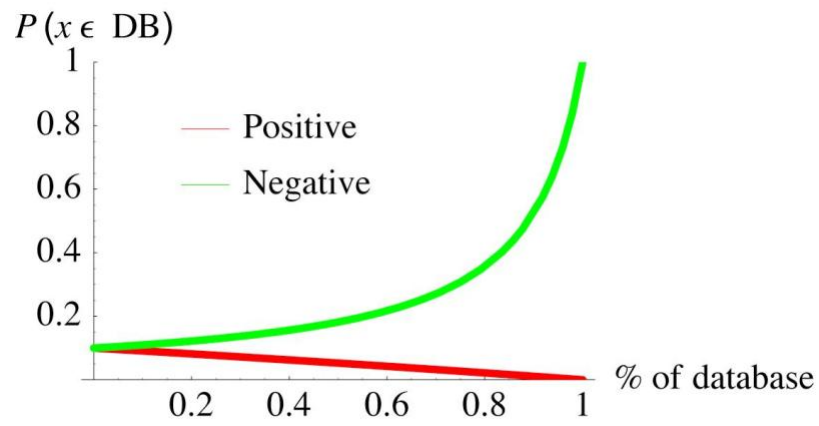


Figure 6.1: DB contains 10% of all possible L-length strings (formulas)

Chapter 7

Use Case Diagram

We need to give the test document to the trained neural network, whose noun phrases are also represented in the form of feature vectors using the similar method applied on the training documents, see chapters 6 and 7 and figure 4. We save the output produced by the classifier for each test document in a separate file. Then we rank the phrases using the algorithm shown in chapter 7 for keyphrase extraction. After ranking the noun phrases, K- top ranked noun phrases are selected as keyphrases for each input test document. A programming environment can be realized by using JAVA.

All the modules of the project framework are already discussed in detail. Now its the time to check the implementation feasibility. We need to explore the system specifications required both in terms of hardware and software. After that, the testing should be done with real world examples. Each stages of the extractor are tested and the achieved results are evaluated.

Chapter 8

State Transition Diagram

The proposed method for keyphrase extraction was described and implemented. The method uses neural network for training and classification. Candidate phrases are giving weights depending on their features and these are given to the network for training and testing. An algorithm is designed for selecting the K top ranked phrases as keyphrases. Authors are giving the right to determine the number of keyphrases, say K. The work is extensible and modifiable. As I mentioned in the abstract, it is possible to include or exclude the features, which determine the results. We can include other structural features of the phrases for producing more accurate results like (1) Appears in the title or headings (2) Distribution in the paragraphs of a given document (3) First occurrence (4) Format of the word (Italic, underlined, bold etc.) (5) Contain special characters (6) Wikipedia References etc. This inclusion may require the input documents as PDF or in word format. Planning to give more emphasis on POS taggers, which can refine the result more. The framework needs to be extended to other domains also.

Chapter 9

Database Schema

The proposed method for keyphrase extraction was described and implemented. The method uses neural network for training and classification. Candidate phrases are giving weights depending on their features and these are given to the network for training and testing. An algorithm is designed for selecting the K top ranked phrases as keyphrases. Authors are giving the right to determine the number of keyphrases, say K. The work is extensible and modifiable. As I mentioned in the abstract, it is possible to include or exclude the features, which determine the results. We can include other structural features of the phrases for producing more accurate results like (1) Appears in the title or headings (2) Distribution in the paragraphs of a given document (3) First occurrence (4) Format of the word (Italic, underlined, bold etc.) (5) Contain special characters (6) Wikipedia References etc. This inclusion may require the input documents as PDF or in word format. Planning to give more emphasis on POS taggers, which can refine the result more. The framework needs to be extended to other domains also.

Chapter 10

Implementation

The proposed method for keyphrase extraction was described and implemented. The method uses neural network for training and classification. Candidate phrases are giving weights depending on their features and these are given to the network for training and testing. An algorithm is designed for selecting the K top ranked phrases as keyphrases. Authors are giving the right to determine the number of keyphrases, say K. The work is extensible and modifiable. As I mentioned in the abstract, it is possible to include or exclude the features, which determine the results. We can include other structural features of the phrases for producing more accurate results like (1) Appears in the title or headings (2) Distribution in the paragraphs of a given document (3) First occurrence (4) Format of the word (Italic, underlined, bold etc.) (5) Contain special characters (6) Wikipedia References etc. This inclusion may require the input documents as PDF or in word format. Planning to give more emphasis on POS taggers, which can refine the result more. The framework needs to be extended to other domains also.

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Appendix A

System Requirements and Frameworks

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A.1 Heading 1

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A.2 Heading 2

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Appendix B

Sample Code

Listing B.1: Code for handling registration

```
#!/usr/bin/perl
use strict;
use warnings;
for (1..99) { print $_."Luftballons\n"; }
# This is a commented line
my $string = "Hello_World!";
print $string."\n\n";
$string =~ s/Hello/Goodbye/;
print $string."\n\n";
test();
exit;
sub test { print "All_good.\n"; }
```

Listing B.2: Code for finding out the errors

```
#include<stdio.h>
#include<semaphore.h>
#include<pthread.h>
#define N 5
#define left (N+n-1)%N
#define right n%N
sem_t s[N];
pthread_t t[N];
void *philo(int n)
{
    while(1)
    {
        think(n);
        sem_wait(&s[left]);
        sem_wait(&s[right]);
        eat(n);
        sem_post(&s[left]);
        sem_post(&s[right]);
    }
}
```

```
}  
void eat(int n)  
{  
    printf("Philosopher %d is eating %d",n);  
    sleep(1);  
}  
void think(int n)  
{  
    printf("thinking %d",n);  
    sleep(1);  
}  
main()  
{  
    int i;  
    for (i=0;i<N;i++)  
        sem_init(&s[i],0,1);  
    for (i=0;i<N;i++)  
        pthread_create(&t[i],0,(void *)  
            philo,(void *)i);  
    while(1);  
}
```

Appendix C

Screen Shots

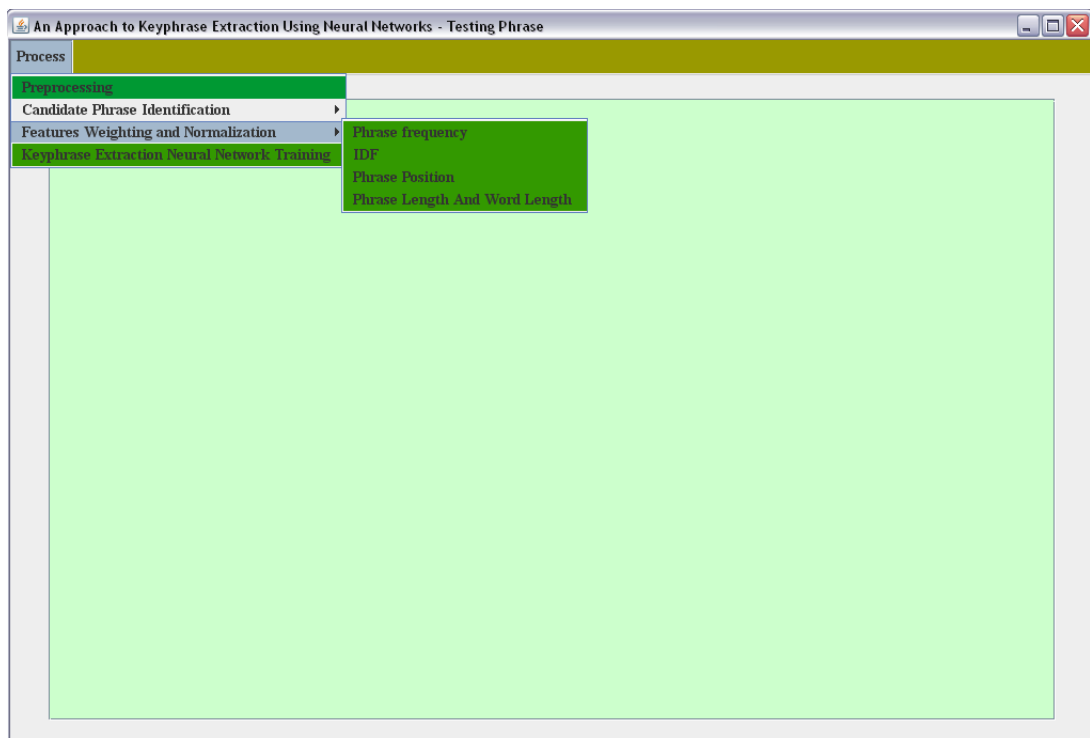


Figure C.1: DB contains 10% of all possible L-length strings (formulas)

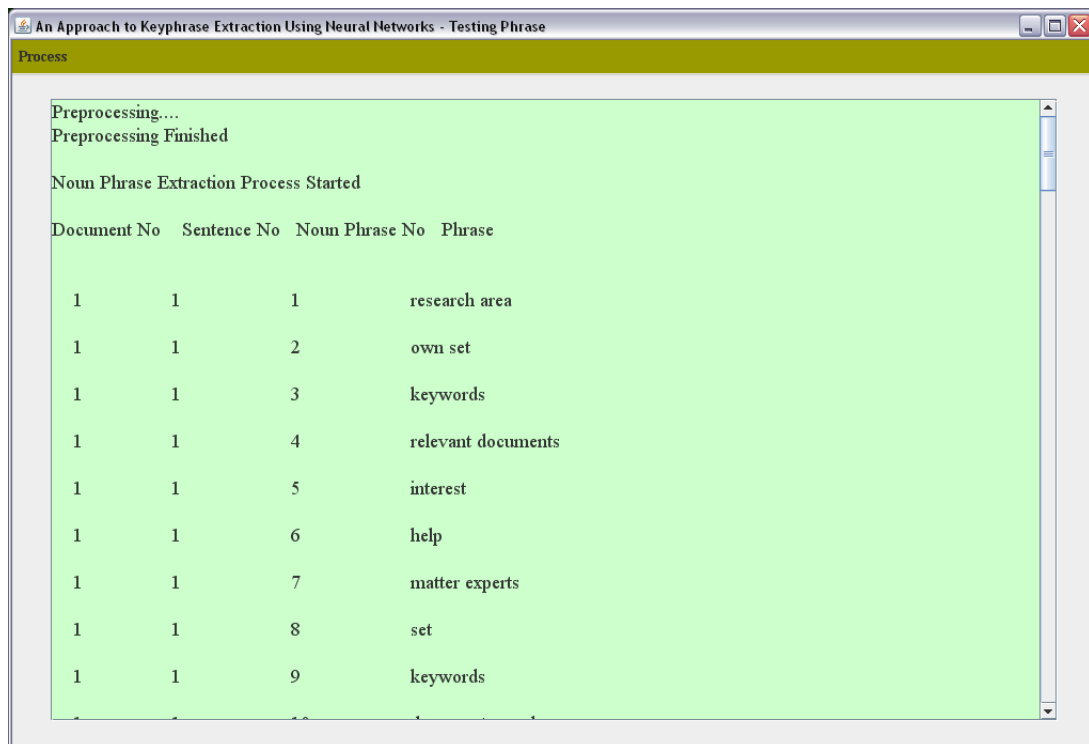


Figure C.2: DB contains 10% of all possible L-length strings (formulas)

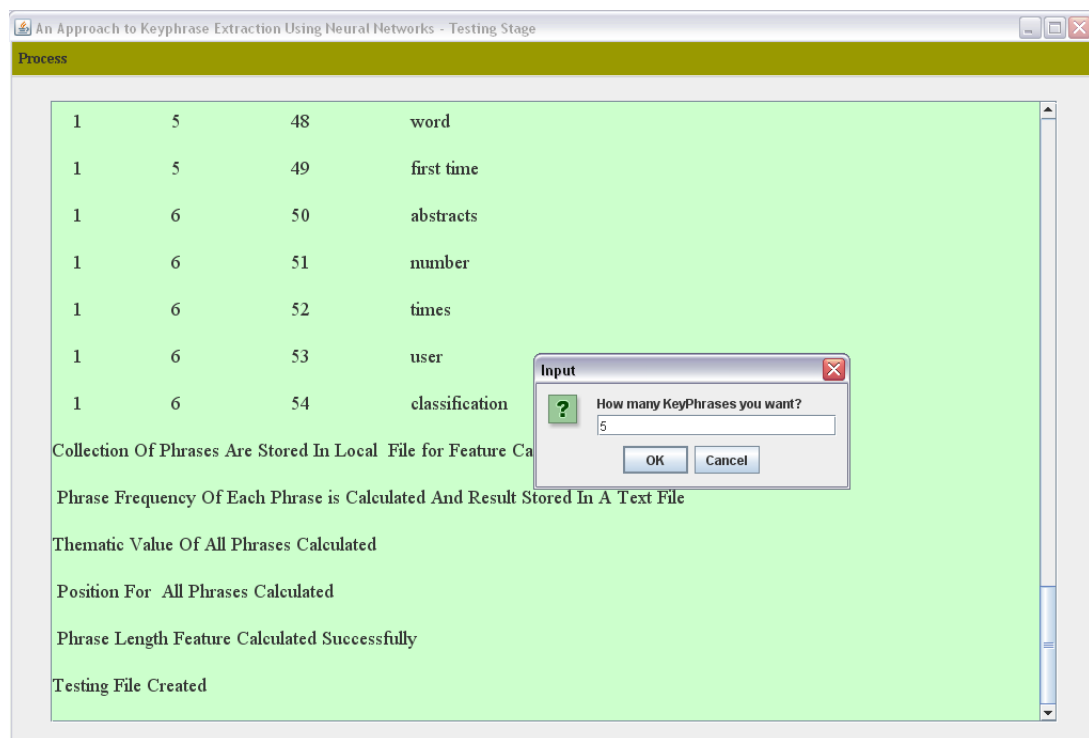
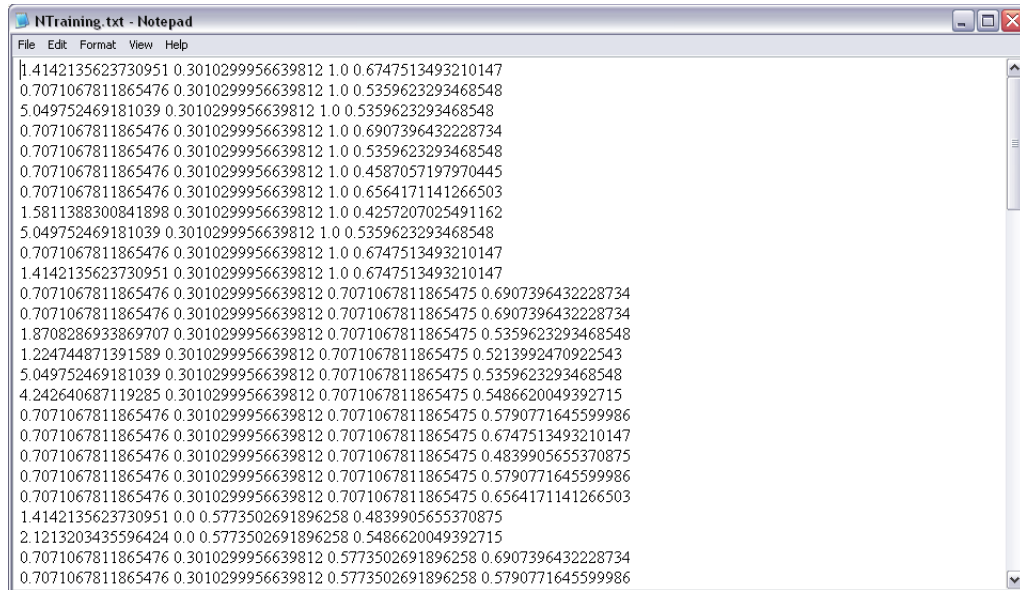


Figure C.3: DB contains 10% of all possible L-length strings (formulas)



```

NTraining.txt - Notepad
File Edit Format View Help

1.4142135623730951 0.3010299956639812 1.0 0.6747513493210147
0.7071067811865476 0.3010299956639812 1.0 0.5359623293468548
5.049752469181039 0.3010299956639812 1.0 0.5359623293468548
0.7071067811865476 0.3010299956639812 1.0 0.6907396432228734
0.7071067811865476 0.3010299956639812 1.0 0.5359623293468548
0.7071067811865476 0.3010299956639812 1.0 0.4587057197970445
0.7071067811865476 0.3010299956639812 1.0 0.6564171141266503
1.5811388300841898 0.3010299956639812 1.0 0.4257207025491162
5.049752469181039 0.3010299956639812 1.0 0.5359623293468548
0.7071067811865476 0.3010299956639812 1.0 0.6747513493210147
1.4142135623730951 0.3010299956639812 1.0 0.6747513493210147
0.7071067811865476 0.3010299956639812 0.7071067811865475 0.6907396432228734
0.7071067811865476 0.3010299956639812 0.7071067811865475 0.6907396432228734
1.8708286933869707 0.3010299956639812 0.7071067811865475 0.5359623293468548
1.224744871391589 0.3010299956639812 0.7071067811865475 0.5213992470922543
5.049752469181039 0.3010299956639812 0.7071067811865475 0.5359623293468548
4.242640687119285 0.3010299956639812 0.7071067811865475 0.5486620049392715
0.7071067811865476 0.3010299956639812 0.7071067811865475 0.5790771645599986
0.7071067811865476 0.3010299956639812 0.7071067811865475 0.6747513493210147
0.7071067811865476 0.3010299956639812 0.7071067811865475 0.4839905655370875
0.7071067811865476 0.3010299956639812 0.7071067811865475 0.5790771645599986
0.7071067811865476 0.3010299956639812 0.7071067811865475 0.6564171141266503
1.4142135623730951 0.0 0.5773502691896258 0.4839905655370875
2.1213203435596424 0.0 0.5773502691896258 0.5486620049392715
0.7071067811865476 0.3010299956639812 0.5773502691896258 0.6907396432228734
0.7071067811865476 0.3010299956639812 0.5773502691896258 0.5790771645599986

```

Figure C.4: DB contains 10% of all possible L-length strings (formulas)



```

An Approach to Keyphrase Extraction Using Neural Networks - Testing Stage
Process

Collection Of Phrases Are Stored In Local File for Feature Calculation

Phrase Frequency Of Each Phrase is Calculated And Result Stored In A Text File

Thematic Value Of All Phrases Calculated

Position For All Phrases Calculated

Phrase Length Feature Calculated Successfully

Testing File Created

---Keyphrases---
phrases
Bengali&documents
keyphrases
document
study
The Keyphrases are Extracted Successfully
You can try to get another phrases by changing the options

NO Of Correct Findings: 2.0
NO Of Author Assigned Data 2
NO Of Extracted Data 5
Precision Value: 1.0
Recall Value: 0.4

```

Figure C.5: DB contains 10% of all possible L-length strings (formulas)